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The Social Consequence of Bureaucratic Oversight: Evidence From the Great Chinese Famine

Ning He

University of Georgia, Athens, Georgia, USA

Correspondence: Ning He (ning.he@uga.edu)

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ABSTRACT

Existing literature shows that increasing political oversight of bureaucrats can improve the quality of government service delivery. Yet when the state is mainly concerned with maximizing revenue extraction from society, increasing bureaucratic oversight may result in social loss. I illustrate this argument with evidence from the Great Chinese Famine, drawing on panel data covering over 2000 counties. I show that weather shocks, which increased central-local information asymmetry on local grain production, reduced the central state's capability to effectively monitor local bureaucrats' effort in grain extraction. This informational barrier to oversight led to greater local autonomy in setting grain extraction quotas. During the famine, this autonomy allowed county officials to relax the execution of excessive grain extraction targets from above, thereby reducing the mortality costs of the state's grain extraction policy. This finding highlights the perils of bureaucratic control in authoritarian states.

1 | Introduction

Bureaucrats are positioned at the forefront of policy implementation. The effectiveness of bureaucrats in implementing state policies has a critical bearing on citizens' welfare (Besley et al. 2022; Grossman and Slough 2022; Pepinsky et al. 2017). While growing literature shows that greater bureaucratic oversight–the degree of a principal's control over bureaucrats' behavior–can increase bureaucrats' compliance in executing policies, this compliance may have varied impacts on citizens (Brierley 2020; Gulzar and Pasquale 2017; Raffler 2022; Slough 2024). When state policies are aligned with citizens' preferences, increasing bureaucratic oversight can improve the quality of policy implementation and thus benefit citizens. However, state policies may not consistently reflect citizens' preferences, especially in contexts where citizens have limited institutional means to hold political elites accountable. In many authoritarian states, the ruling elite faces minimal institutional constraints in extracting the maximum attainable social surplus from society (Olson 1993). When authoritarian state leaders prioritize maximizing extraction, how does bureaucratic oversight affect its implementation and shape citizens' welfare?

This article studies the social consequences of bureaucratic oversight in authoritarian states. I argue that when authoritarian state leaders have limited capability to monitor bureaucrats, bureaucrats exert less effort in executing unpopular state policies, potentially mitigating the social costs of the policies. I illustrate this argument by examining how barriers to monitoring due to information asymmetry affects bureaucratic behavior in revenue extraction, building on recent literature that investigates the relationship between information asymmetry and the development of state hierarchies (Garfias and Sellars 2024; Lee and Zhang 2017; Mayshar et al. 2017). I argue that effective monitoring by the principal constrains bureaucratic discretion, leading to closer adherence to the directives from the principal in policy implementation. Conversely, when the principal lacks full knowledge of bureaucratic effort, the information gap creates space for greater bureaucratic discretion, potentially leading to more adaptable policy implementation that fits local conditions. While such autonomy of bureaucrats may hamper "bureaucratic accountability" to authoritarian state leaders, it allows for greater responsiveness to citizens through the exercise of discretionary power. When executing state policies bearing potential costs for society, bureaucrats with greater discretion can take measures to moderate the policies' negative impacts on citizens.

I test this argument using the Great Chinese Famine between 1959 and 1961 as an empirical context. In 1958, Chinese Communist Party (CCP) leader Mao Zedong launched the so-called Great Leap Forward (GLF) campaign with the goal of rapidly increasing China's agricultural output. Local government officials were pressured to boost grain production and significantly increase grain extraction from rural communities. Driven by political pressure from above, local officials extracted excessive amounts of grain from rural communities, resulting in severe food shortages in rural areas. The over-extraction is recognized as one of the main causes of the famine between 1959 and 1961, which claimed the lives of over 30 million people. Moreover, mortality rates in the famine varied significantly across regions, partly due to local officials' varying effort in fulfilling state grain extraction targets.

Drawing on new county-level data, I test how information asymmetry between county officials and their superiors impacted famine mortality by shaping the implementation of state grain extraction targets in counties. I use weather shocks as a proxy for information asymmetry and the state's capacity for bureaucratic oversight. I show that weather shocks caused unforeseen fluctuations in county grain output, which reduced higher-level officials' ability to estimate county grain production, thus increasing barriers for them to monitor county officials' effort in grain extraction. This information asymmetry afforded county officials greater independent authority in setting county grain extraction quotas. During the famine, this discretion allowed them to relax the execution of excessive grain extraction targets from above, thereby mitigating the mortality costs of the CCP's grain extraction policy. I find that a one-unit increase in the deviation of precipitation from the long-term average was associated with a 6% decrease in famine mortality rates. This pattern was explained by grain extraction: during the famine, farmers' grain retention level was significantly higher in counties that experienced weather shocks.

I further show that the impact of weather shocks was correlated with the social proximity of county officials to the county population they governed. Combining a dataset on county party secretaries with language dialect data, I construct a measure for bureaucratic social proximity based on the degree of similarity between the dialect spoken by a county party secretary and the dominant dialect spoken by the county population. I show that weather shocks were associated with a greater decrease in famine mortality when a low dialect distance was observed. In contrast, the effect of weather shocks was minimal when the county party secretary spoke a dialect that was significantly distinctive from the local dialect. These heterogeneous effects were likely driven by county party secretaries' varying responsiveness to local citizens: socially proximate party secretaries were more inclined to contain the social costs of excessive state targets in grain extraction using their discretionary power.

This article extends a growing body of research that investigates the political logic of bureaucratic control in authoritarian states. Previous works show how autocrats strategically manage bureaucrats to counter internal threats to strengthen their hold on power (Greitens 2016; Hassan 2020; Montagnes and Wolton 2019), yet there has been little research on how bureaucratic control by autocrats affects the well-being of ordinary citizens. This article demonstrates that when citizens are unable to hold political leaders accountable, greater state capacity to enforce policy agenda through a compliant bureaucracy can bring significant costs to the population. The finding provides a needed synthesis to existing theories on the causes of social disasters under authoritarian regimes, which explain social disasters from the separate lenses of political accountability (e.g., Sen 1999) and state capacity (e.g., Scott 1998).

Second, this article extends the literature on how information asymmetry shapes institutional development. A growing body of research highlights the role of information asymmetry in shaping the institutional paths of historical states (Ahmed and Stasavage 2020; Garfias and Sellars 2024; Mayshar et al. 2017). While the adoption of centralized bureaucracy is often suggested as a solution to the information problem in strengthening state extractive capacity, this perspective overlooks the agency problems between rulers and bureaucrats. The consequences of such agency problems for revenue extraction–especially their variations within a country–are not well understood. This article shows that the variations in both local conditions and bureaucratic characteristics can meaningfully influence how these agency problems play out, leading to divergent social outcomes even under uniform institutions and political mandates.

Third, this article also contributes to our understanding of the causes of famine. Contrary to the view that the Great Chinese Famine was a consequence of "three consecutive years of bad weather," this article shows that weather shocks moderated the negative social impact of state grain extraction policy during the famine. When local officials were compelled to pursue excessive grain extraction targets, weather shocks gave the sympathetic officials an excuse to lessen state extraction targets, potentially saving numerous lives. This finding reveals a nuanced relationship between weather and famine mortality. The implication aligns with classic works on the institutional origins of mass starvation in human societies (Sen 1981, 1999). Relative to weather's direct impact on agricultural output, how the government allocates limited resources in response to weather shocks can be fundamental in shaping the weather-to-death relationship.

Finally, this study contributes to the literature on the political economy of bureaucracy in China. Much of the literature emphasizes bureaucratic incentives in shaping revenue extraction (He and Wu 2023; Kung and Chen 2011; Z. Li and Yu 2023), which provides a basic framework for understanding bureaucratic compliance in pursuing state targets. This study shifts attention to when and why bureaucrats deviate from central mandates, focusing on the variations in hierarchical oversight and social embeddedness of bureaucrats. The finding explains not only bureaucratic compliance but also conditional divergence, enriching our understanding of bureaucratic discretion and local agency in authoritarian governance.

2 | Bureaucratic Discretion Under Information Asymmetry

2.1 | Existing Research

Information plays a critical role in shaping the principal-agent relationship in bureaucracies. Bureaucrats possess policy-relevant information that is essential for the principal–elected officials or a central government–to monitor their effort. However, such information is often not accessible to the principal. This information asymmetry constrains the principal's capability to monitor and sanction bureaucrats, which further affects the degree of bureaucratic discretion and policy implementation outcomes.

Information asymmetry increases the extent of autonomy and discretion enjoyed by bureaucrats in policy implementation, as demonstrated by recent scholarship on state building. Analyzing the principal-agent relationship between a central elite and local agents, studies show a negative correlation between local autonomy and the central elite's capability to accurately estimate local production levels (Garfias and Sellars 2024; Mayshar et al. 2017). Greater transparency of local production exposes local agents to increasing scrutiny by the central elite, allowing for closer monitoring and potential sanctions. This restricts the agent's autonomy in carrying out their duties, leading them to adhere more closely to the elite's directives in taxing local production. Conversely, lower transparency of local production increases barriers for the central elite to monitor the agent's effort, which affords the agent greater independent authority in taxation. Limited transparency of local conditions can further increase the likelihood of moral hazard actions, where the bureaucrat implements a tax rate that deviates from the optimal level for the elite.[1]

The autonomy of bureaucrats in implementing state policies has heterogeneous downstream impacts, depending on the incentives of bureaucrats and their principal. When bureaucratic autonomy is low, the principal's incentives play a greater role in shaping policy outcomes. Such incentives are linked to institutions that structure the accountability relationship between politicians and citizens (Besley and Burgess 2002; Sen 1999). Elected officials may lack incentives to motivate bureaucrats when they find it difficult to claim credit for the effort in elections (Gulzar and Pasquale 2017). Furthermore, politicians can take advantage of their discretionary authority over bureaucrats to extract rent when political accountability is weak (Brierley 2020). In authoritarian states, the ruler can strategically post and shuffle bureaucrats to maximize their compliance in carrying out state repression (Hassan 2020), yet they have limited interest in motivating bureaucrats for better public goods provision (Mesquita et al. 2005; Egorov et al. 2009).

Greater bureaucratic autonomy allows bureaucrats' own incentives to drive policy outcomes beyond those set by the principal. Apart from official rules set by the principal to regulate awards and penalties for bureaucrats, informal institutions–like bureaucrats' social relationships–also play a role in shaping their behavior and performance (Bhavnani and Lee 2018; Bozcaga 2020; Jiang 2018). Socially embedded bureaucrats face greater community pressure that helps ensure their responsiveness to citizens in service delivery, which may compensate for the absence of electoral accountability (Tsai 2007). Social proximity between bureaucrats and local citizens has a critical impact on government responsiveness to citizens during social crises: it provides incentives for bureaucrats to address citizens' urgent needs (G. Xu 2021).

2.2 | Theoretical Framework

To illustrate how information asymmetry shapes bureaucratic actions in revenue extraction, consider a simple principal-agent model. A local bureaucrat, acting as the agent of a central elite, oversees grain extraction from the local population. Accountability in this delegation depends on the elite's ability to monitor the bureaucrat's effort. The monitoring capacity ensures that shirking can be detected and punished. However, delegation inherently generates monitoring problems. The elite cannot directly observe the bureaucrat's work and relies instead on their output, such as the amount of grain extracted. The level of extraction can be a noisy signal of extraction effort because it is also affected by local conditions like weather. The mapping between extraction and effort is nondeterministic; rather, it contains a random component beyond the bureaucrat's control.

Given the noisy relationship between realized extraction and a bureaucrat's effort, the level of local grain output provides crucial information for the principal to infer the agent's effort. When extraction is set as a deterministic function of total output (e.g., a certain percentage), the agent's effort can be inferred based on output and extraction. For illustration, let $p \in \{0, 1\}$ represent grain production outcome, where $p = 1$ denotes normal production and $p = 0$ represents poor production. The agent's effort in extracting the production is denoted by $e \in \{0, 1\}$, where $e = 1$ represents high effort and $e = 0$ represents low effort. Extraction outcome is represented by $pe \in \{0, 1\}$. The principal cannot observe p or e in separation, but only the extraction outcome pe. When $pe = 1$, the quantity of extraction meets the elite's expectation, which is observed only when production is normal, and the bureaucrat exerts high effort. In this situation, the agent's effort level can be easily inferred. However, when extraction falls short of expectation, whether the principal can accurately infer the agent's effort level depends on their knowledge of grain production. The quality of this knowledge reflects the legibility of local grain output to the central elite.

The legibility of local conditions is shaped by a range of factors, such as geography and technology (Garfias and Sellars 2024; Mayshar et al. 2017). It is also shaped by state actions to increase administrative capacity for the purpose of monitoring bureaucrats (Ma and Rubin 2019). Here I consider an exogenous factor: weather shocks. In agrarian societies dependent on rain-fed agriculture, production hinges on precipitation patterns. Consistent and normal rainfall allows the central elite to establish a baseline for expected output, which can facilitate monitoring of local bureaucrats. In contrast, deviations from normal precipitation levels lead to unpredictable fluctuations in grain output. When the elite relies solely on local bureaucrats to collect information about the output, these unpredictable fluctuations exacerbate the elite's information problem by creating opportunities for bureaucrats to misreport. The elite's knowledge of local production is thus a function of local weather conditions. The elite faces greater uncertainties about local grain production under weather shocks, compared with under normal weather conditions.

Upon observing the quantity of grain extracted by the bureaucrat and local weather conditions, the elite decides whether to retain or dismiss the bureaucrat. A bureaucrat will not be dismissed if the quantity of extracted grain meets the elite's expectation, regardless of the weather. When grain extraction falls short of expectation, however, the elite's decision to dismiss an agent will be affected by weather conditions. When precipitation is abnormal, the elite is unable to distinguish if low extraction is due to poor effort or decline in grain production, increasing the likelihood of wrongful dismissal that would disincentivize other bureaucrats from exerting maximum effort. The principal can avoid this cost by choosing an arrangement where the agent is allowed to independently decide the extraction level without risking dismissal, which corresponds to a "pure-carrot" contract between the elite and the bureaucrat (Mayshar et al. 2017). The elite's incentive to grant such authority to the agent diminishes as legibility increases (e.g., precipitation is normal), since they can better distinguish between shirking agents and those facing legitimate production difficulties.

The bureaucrat exerts a certain level of effort that maximizes their utility. Beyond pecuniary benefits from staying in the position and disutility from exerting effort, their utility function may also include the potential cost of their grain extraction actions on society. Whether or not this cost factors into a bureaucrat's utility function depends on social proximity between bureaucrats and local citizens. Socially proximate bureaucrats are more likely to internalize the potential negative impacts of extraction on local citizens. While a socially distant bureaucrat may consistently exert high effort in extracting local production, a socially proximate bureaucrat exerts high effort only under normal weather conditions.

The above discussion can be summarized as the following intuitions: In a principal-agent relationship characterized by information asymmetry, the principal is expected to delegate greater independent authority to the agent. The agent, as a result, is more likely to make discretionary adjustments in policy implementation. This delegation arrangement can impact both bureaucratic accountability to the principal and their responsiveness to citizens, conditional on the social proximity of bureaucrats to local citizens.

3 | Empirical Context and Hypotheses

3.1 | Grain Extraction Under Mao

After its victory in the Chinese Civil War, the CCP conducted a series of political campaigns to gain greater control over rural society and its economic production. By 1952, the CCP had swept through most rural communities with land reform, which eliminated the land-owning elite and established local networks of the party-state (Schurmann 1966, Chapter 7). The party also tightened its grip on agricultural output by closing free grain markets and forcing farmers to sell their grain only to the state. The collectivization of agriculture since 1953 further eliminated individual land ownership. The land and its associated production were held as collective assets, leaving individual peasants no control over their production. State officials dictated what qualified as surplus grain and forced rural collectives to sell a vast majority of the surplus grain to the state. Grain rations allocated to peasants, also determined by state officials, were insufficient by most standards and frequently encroached upon by state procurement (Oi 1989).

State grain procurement was implemented under a top-down planning system. Targets for grain procurement were set at the provincial level in proportion to the projected grain output, which was based upon historical production records. These targets then cascaded down the administrative hierarchy and were divided among governments at lower levels (Oi 1989, Table 10). In 1955, a quota system was introduced to regulate how local officials set grain procurement targets. The central government mandated that local districts fix their grain procurement quotas for three consecutive years, basing these quotas on the grain production level observed at the beginning of the 3-year period. The fixed quotas served two purposes: as state targets for grain extraction and as benchmarks for evaluating local officials. Officials at each level faced immense pressure to fulfill those quotas, risking punishment for shortfalls.

The fixed grain procurement quotas could be too rigid for implementation. They required that grain production levels did not fluctuate much from year to year. Adjustments to the quotas were needed when the grain production level fell significantly from the benchmark year so that peasants' grain rations were not encroached upon. In accommodating such situations, local officials were allowed to adjust the quotas in response to unforeseen circumstances like weather shocks. This discretion for local officials was acknowledged by the central government in a number of policy documents (see Supporting Information S1: A1) and also noted by historians (e.g., Bernstein 1984). In guiding the adjustments, the central government discouraged local officials from making significant reductions to the state's share and insisted on the priority of the established target. The actual implementation, however, was discretionary and exhibited variations across regions. For example, as Walker (1984, 48) observed from official statistics, while grain output fell due to natural disasters in both Anhui and Hebei in the 1950s, the degree of adjustments in grain taxes following the disasters exhibited a notable variation between these provinces.

A notable change in the state's grain procurement targets was observed during the GLF. The campaign was initiated by Mao with the goal of rapidly increasing the country's agricultural output through organizational changes and political mobilization (Walder 2015, 154). The GLF was set in motion with decentralization of economic planning, as well as political pressure on local officials to increase grain extraction from rural collectives. To echo Mao's call for a great leap forward in the agricultural sector, local officials experimented with a variety of misguided production measures aimed at increasing grain output, which nonetheless turned out to be counterproductive and led to declines in grain output (Walder 2015, 160–161). Despite the looming failures, local officials competed to report falsely exaggerated production successes and wildly inflated statistics on output, which were then used to determine grain procurement quotas (H. Xu and Tian 2020). The campaign to exaggerate grain output, also known as announcing "grain satellites," amplified the tension between the state's agenda and the welfare of local population, which created a dilemma for local bureaucrats. While fabricating crop yields signaled adherence to the political line championed by Mao, it also meant that a larger amount of grain had to be extracted from rural collectives. Nationwide grain procurement increased by 40% from 1957 to 1959 while output fell by 13% (W. Li and Yang 2005). The decline in production, combined with the expanding grain procurement quotas, resulted in excessive grain extraction that cut into the food rations for rural residents. This is recognized as one of the main causes of the famine of 1959–1961 (Bernstein 1984), during which over 30 million people perished (Cao 2005).

3.2 | Hypotheses

The information asymmetry that emerged in grain procurement process has been well noted by scholars (Kung 2022, 665, Meng et al. 2015, 1595, Oi 1989, 124). Limited access to real-time data on grain output forced higher-level officials to rely on estimates when setting grain procurement targets and to base their evaluations of lower-level officials' performance in grain procurement on those estimates. When weather conditions were normal, higher-level officials expected a certain level of grain output based on historical production records. With weather shocks, however, local grain output became more variable and less predictable. The fluctuations in grain output widened the information gap between local officials and their principals at upper levels, creating opportunities for lower-level officials to adjust procurement quotas following their own judgment of the situation. This discretionary adjustment under weather shocks can be observed as increasing variations in grain retention levels, which reflected food availability in rural communities after state grain extraction.

Hypothesis 1. *In counties affected by weather shocks, there were higher variations in grain retention.*

The discretion associated with weather shocks may have resulted in less excessive grain extraction during the GLF. When upper-level officials lacked the ability to personally observe the actual output, lower-level officials could exaggerate the production decline caused by adverse weather. Consequently, actual grain procurement might deviate from the principle of prioritizing the state's share, thus allowing farmers to keep more grain than what would have been considered to be optimal by the state absent information asymmetry. In this situation, the production cost of bad weather was allocated more toward the state than toward the rural population. Such discretionary adjustments might have significant social ramifications during the famine, when rural population's survival hung precariously on the residuals of state procurement.

In the context of the GLF, adverse weather conditions made achieving exceptional production records more challenging, causing even hardworking officials to fall behind in meeting the grain procurement targets. Local officials who failed to extract as much grain as expected could potentially escape penalty in the presence of weather shocks. Thus, weather shocks relieved some local officials–especially those who were concerned about the potential social cost of excessive grain extraction–from the pressure to set unrealistic state targets (e.g., announcing grain satellites). This might further contribute to the variations in grain retention and famine mortality across regions.

Hypothesis 2. *In counties affected by weather shocks, officials extracted grain less excessively and left more grain in rural communities during the famine, which resulted in lower famine mortality.*

While weather shocks could have provided greater discretion for local officials, how they used this discretion depended on their level of social proximity. Socially proximate officials were more likely to be responsive to the local population. They were more inclined to increase grain retention in rural communities by taking advantage of their discretionary power. In contrast, socially distant officials were less likely to do so despite their discretionary power. Consequently, the effects of weather shocks are expected to vary based on bureaucratic social proximity. Weather shocks were likely to mitigate famine severity primarily in localities where officials were socially proximate to the local population.

Hypothesis 3. *Weather shocks had a larger effect on reducing grain extraction and famine mortality when a county party secretary was socially proximate to the county population.*

4 | Data

I assemble a dataset in panel structure, which covers 2218 county-level administrative units for the period from 1954 to 1965. County officials responded directly to the principals at the prefecture level, while authorities further above–provincial and central authorities–acted as indirect principals. At all levels of the administrative hierarchy, however, weather shocks may create potential information asymmetries regarding local grain production. Furthermore, county officials played a significant role in executing state grain procurement policy within their jurisdictions, with considerable authority in determining the procurement quotas (Oi 1989, 41). The intensity of grain extraction and famine mortality rates exhibited substantial variations across counties, making these administrative units well-suited for examining the role of bureaucratic behavior in the famine.

4.1 | Death Rates

To measure famine severity, I collect county-level death rate data from the *Population Statistical Material* for each province, spanning the period from 1954 to 1965. These data are supplemented by death rate data from county gazetteers when the statistical material is not available for a province. For each county-year observation, death rate is defined as the number of deaths per thousand population. In ordinary times, the average death rate was between 11 and 14. During 1959–1961, it reached 18, 28, and 16 respectively. Famine death rates varied substantially across counties (Figure 1), with a standard deviation of 26 in 1960.

To address concerns about potential manipulation and censorship of the death rate data, I adopt a second measure for famine severity. I use a random sample from the 2000 population census and calculate population size for each birth cohort by county. Compared with the death rate data from official statistical materials, the census data are less susceptible to manipulation. The combined effect of famine on fertility and infant mortality resulted in significant declines in population born during the famine years, which was 30–40% smaller than other birth cohorts.

4.2 | Weather Shocks

I construct the measure for weather shocks using precipitation data, which are digitized from *China Precipitation Materials* compiled by the Central Meteorological Bureau. These materials provide annual data on precipitation levels between 1951 and 1970, observed at over 450 locations evenly distributed across the country. To estimate precipitation at the county level, I use spatial interpolation to predict the values for each county. This generates a county-year panel of precipitation observed for 2 decades.[2] I use this panel to construct the measure for weather shocks, which reflects the degree that the precipitation level in a county and year deviated from this county's long-term average. The calculation for this measure can be expressed in the following equation:

$$\mathrm{RD}_{\mathrm{it}} = \frac{|R_{it} - \bar{R}_i|}{\sigma(R)_i}$$

where RD_{it} is *rainfall deviation* in county i and year t. It is a function of the gap between the precipitation level in county i and year t, R_{it}, and the average precipitation level in county i over the period between 1951 and 1970, $\bar{R}_i$, then divided by the standard deviation of this county's precipitation over the same time, $\sigma(R)_i$. I employ the absolute value as the measurement of weather shocks, which captures the magnitude of rainfall deviation in both directions.

The overall weather conditions across the country, as reflected by county-level rainfall deviation, was not significantly different during the famine in comparison to ordinary years (see Supporting Information S1: A2). The proportion of counties with abnormal precipitation was similar between famine and non-famine years, casting doubt on the claim that the famine was caused primarily by adverse weather.

Figure 2 illustrates the relationship between rainfall deviation and death rates. The left panel, based on data from 1956, shows no distinguishable correlation between rainfall deviation and death rates, suggesting that pre-famine mortality was not quite sensitive to weather conditions. In non-famine years, farmers likely retained adequate grain for survival regardless of weather conditions. As a result, mortality exhibited limited variations and remained uncorrelated with rainfall deviation. A different pattern emerged during the famine. The right panel shows a negative correlation between rainfall deviation and mortality rates in 1960, with lower mortality rates observed in counties that had more abnormal precipitation.

To support a causal interpretation of the estimated effects of weather shocks, I examine whether rainfall deviation during the famine period was correlated with observable county characteristics. Specifically, I regress rainfall deviation from 1959 to 1961 on a range of county characteristics, as shown in Table 1. No significant correlations are found, with the sole exception of longitude. These results confirm that the variation in rainfall deviation was plausibly exogenous.

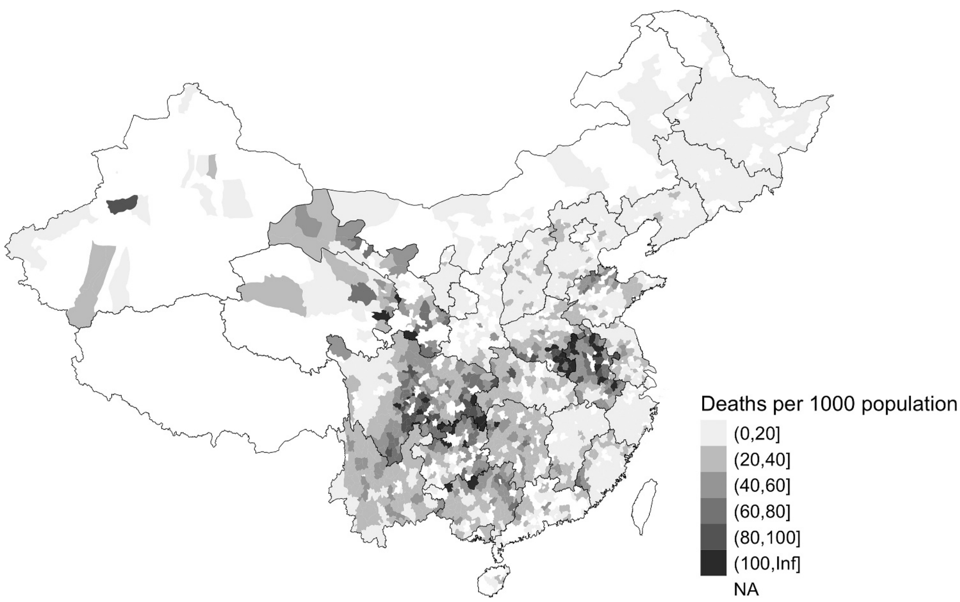

FIGURE 1 | County mortality rate in 1960.

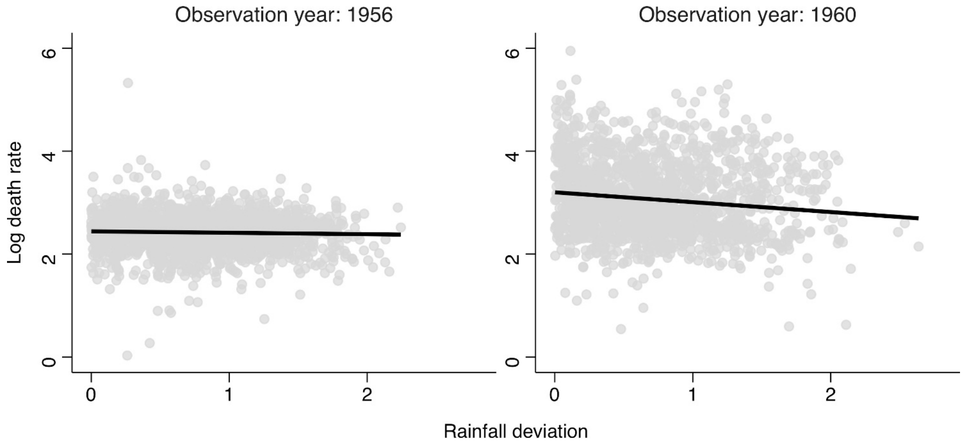

FIGURE 2 | Rainfall deviation and county mortality rate.

TABLE 1 | Balance tests for rainfall deviation.

Variable	Estimate	*p*-value	Observations
Longitude	0.018	0.003	6654
Latitude	0.000	0.998	6654
Terrain ruggedness	0.000	0.124	6654
Average precipitation, 1951–1970	0.000	0.228	6654
Grain productivity, 1957	−0.038	0.067	5499
Rice share, 1957	0.074	0.144	4860
Wheat share, 1957	−0.032	0.716	5328
Population, 1957	0.000	0.155	5655
Purge intensity, 1957	0.000	0.210	6654
Grain procurement rate, 1957	0.001	0.108	3219
Grain retention, 1957	0.000	0.236	3105

Note: Estimated effects of various county characteristics on rainfall deviation between 1959 and 1961. Each coefficient is estimated from a separate regression with one independent variable, while conditioning on province fixed effects and year fixed effects. The *p*-values correspond to standard errors clustered by prefecture.

4.3 | Grain Extraction

To measure grain extraction, I digitize panel data for grain output and state procurement over the period from 1954 to 1965. The data are digitized from multiple sources including statistical materials and county gazetteers. To address potential bias due to missing observations, I restrict my analysis to counties with at least 10 observations for grain output or grain procurement. This results in a nearly balanced panel that covers 1355 counties for grain output and 983 counties for grain procurement. I measure the intensity of grain extraction using grain retention per capita, which is calculated by subtracting state procurement from total grain output, then dividing the difference by county population.[3] Given county officials' authority to set procurement quotas and grain retention levels for rural communities (Oi 1989, 41), this measure captures the discretionary choice made by county leaders when implementing state grain procurement policy. This measure also serves as a proxy for food availability among the population in a county. Figure 3 illustrates how grain retention per capita fluctuated over time. On average, a county resident retained over 200 kg of grain in ordinary years. During the GLF, grain retention per capita plummeted to subsistence levels–around 165 kg per year–a ration barely sufficient for adults whose physical activity was minimal.[4]

4.4 | Dialect Distance

I construct a measure for social proximity between county party secretaries, the chief party leader in a county, and the population they governed. Biographical information for county party secretaries is digitized from county gazetteers, including their time in office and the county in which they were born. This dataset covers the period from 1949 to 1965 for approximately 900 counties, with a total number of 6898 leader terms.

To capture the social proximity of a county party secretary to the citizens in a county, I exploit their dialect distance, a measure of the difference between the official's native dialect and the dominant dialect of the county they governed. The dominant dialect in each county is identified from *The Language Atlas of China* and *The Chinese Dialect Dictionary* (Qin 2022). A county party secretary's native dialect is determined based on their birth county. Following existing research (e.g., Fang and

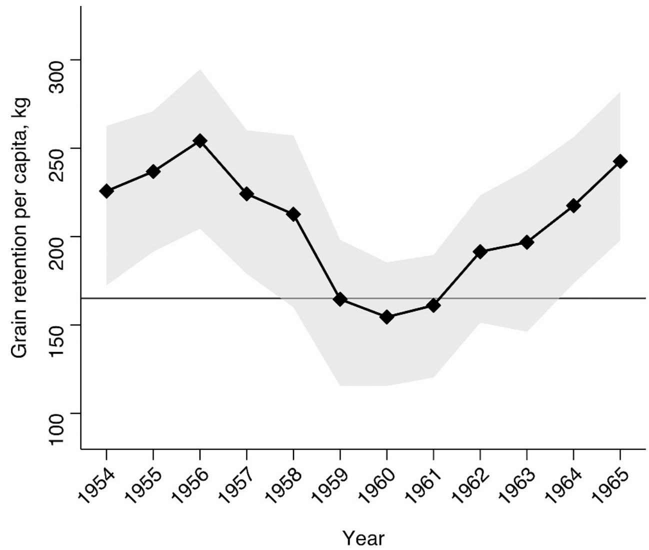

FIGURE 3 | Temporal variations in grain retention. The points represent average grain retention per capita across counties. Intervals are bounded by the 25th and 75th percentiles. The horizontal line marks the subsistence level of grain retention at 165 kg per capita per year.

Hong 2020), dialect distance is determined based on distances between branches on the linguistic tree, as detailed below:

- When two dialects belong to the same variety, the distance is 1. When two dialects belong to the same group but not the same variety, the distance is 2. When two dialects belong to the same family but not the same group, the distance is 3. When two dialects belong to different families, the distance is 4.[5]

The variable ranges from 1 to 4, with lower values indicating a lower dialect distance and higher social proximity. My analysis exploits temporal variations in dialect distance resulting from county party secretaries' turnover. The annual turnover rate ranged from 20 to 30% during the observation period. About half of these turnovers caused a shift in dialect distance (see Supporting Information S1: A4). The average dialect distance was larger in the south than in the north, a pattern that is aligned with historical accounts of cadre appointment in the Mao era (Vogel 1969).

I cross-validate this measurement using a conventional proxy for social proximity: a dummy variable indicating whether a county party secretary was native to the province in which they took office. Dialect distance is strongly correlated with the indicator for native county party secretaries (see Supporting Information S1: A5). Among secretaries with a dialect distance of one, 96% were native to the province, whereas the probability dropped to 19% for those with a dialect distance of four.

5 | Empirical Strategy and Results

This section presents empirical analysis for the impacts of weather shocks. I begin with testing the assumption that weather shocks afforded county officials greater flexibility in the implementation of grain extraction policy. Then I estimate the effect of weather shocks on famine mortality and illustrate the potential mechanism. Next, I show how the effects of weather shocks varied based on the social proximity between county party secretaries and the local population.

5.1 | Weather Shocks and Variations in Grain Retention

Weather shocks could amplify information asymmetry within the bureaucratic hierarchy, making it difficult for higher-level officials to assess local grain output. The CCP's official policy documents show that lower-level officials were granted discretionary authority to adjust procurement quotas in response to unforeseen fluctuations in grain output. It is noteworthy that the local discretion did not arise from the principal's lack of access to information about local weather conditions. In fact, systematic data on precipitation and temperature were already available in the 1950s. The actual information problem lay in the difficulty of predicting the extent of production declines following abnormal weather events. For example, the relationship between rainfall deviation and grain output exhibited substantial variations over time: while rainfall deviation often led to declines in grain output, the magnitude of these declines varied substantially from year to year (see Supporting Information S1: A6).

I use variance function regressions to estimate how weather shocks affected the variances of grain output and grain retention, following the two-stage approach described by Western and Bloome (2009). In the first stage, I fit a linear regression of the outcomes on rainfall deviation and save the model residuals. The second stage is a gamma regression of the squared residuals on rainfall deviation with a log link function.[6] The coefficient of the first stage, which is from a standard linear regression, has the usual interpretation: the average change in the outcomes associated with a one-unit change in rainfall deviation. The coefficient of the second-stage regression, however, captures the impact of rainfall deviation on outcome variances.

Table 2 presents the variance regression estimates for the effects of rainfall deviation on grain output and retention. Columns (1)

TABLE 2 | Variance regressions on the impacts of weather shocks.

	First stage		Second stage	
	Output (1)	Retention (2)	Output (3)	Retention (4)
Rainfall deviation	−0.049***	−0.048***	0.082***	0.128***
	(0.013)	(0.013)	(0.028)	(0.046)
Observations	9392	9392	9392	9392
Adjusted R^2	0.589	0.285		
Year FEs	Yes	Yes	Yes	Yes
Prefecture FEs	Yes	Yes	Yes	Yes

Note: Standard errors clustered by prefecture in parentheses.
***$p < 0.01$.
**$p < 0.05$.
*$p < 0.1$.

and (2) show estimates from the first-stage regressions. The negative coefficient in Column (1) indicates that grain output declined as precipitation deviated from the normal level. Column (2) further shows a negative impact of rainfall deviation on the level of grain retention. Specifically, a one-unit increase in rainfall deviation was associated with a 5% decrease in grain retention. It is noteworthy that the estimated coefficients are similar between Columns (1) and (2). However, since grain output was always higher than grain retention, these coefficients suggest that the absolute decline in grain retention was actually lower than the absolute decline in grain output.[7] This implies that the quantity of procurement was adjusted downward in response to weather shocks, so the production cost of weather shocks was divided between the state and farmers through this adjustment.

Columns (3) and (4) show the results of the second-stage regressions, where the coefficients reflect how the variances of grain output and grain retention changed with rainfall deviation. The estimate in Column (3) shows how rainfall deviation affected the variance of grain output. The positive coefficient indicates that greater rainfall deviation was associated with increasing variability in grain output. This result supports an instrumental assumption underlying the proposed mechanism: precipitation changes had a varying impact on grain output, which became less predictable as rainfall deviation increases. Rainfall deviation thus increased the barrier for higher-level officials to accurately predict how grain output would change. Furthermore, the increasing variance in grain retention following rainfall deviation, as Column (4) shows, suggests that adjustments to the grain procurement quotas were not uniform across counties. County leaders likely exercised greater discretion in setting the procurement quotas during weather shocks.

5.2 | Weather Shocks and Famine Mortality

I estimate the effect of weather shocks on famine mortality with the following specification:

$$y_{it} = \beta_1 \times RD_{it} + \beta_2 \times RD_{it} \times GLF_t + \theta_i + \gamma_t + \varepsilon_{it} \qquad (1)$$

Where the outcome variable y_{it} is the log number of deaths per thousand population in county i and year t. It is a function of rainfall deviation RD_{it}, which is interacted with a year dummy for the famine period (1959–1961). The equation includes county fixed effects θ_i and year fixed effects γ_t, which capture time-invariant confounders across counties and nationwide trends in weather conditions and mortality rates. The interaction coefficient β_2 captures the differential effect of weather shocks on mortality during the famine, also referred to as the effect on famine mortality. A positive estimate of β_2 indicates a larger increase in famine mortality rates for counties experiencing greater rainfall deviation. Conversely, a negative estimate suggests a smaller increase in famine death rates under adverse weather conditions.

Table 3 shows the estimated effect of rainfall deviation on famine mortality. The estimate of β_2, shown in Column (1), is negative and statistically significant. This indicates a lower increase in death rates during the famine for counties experiencing greater rainfall deviation. The point estimate −0.063 suggests that a one-unit increase in rainfall deviation was associated with about 6% decrease in death rates during the famine.[8] Column (2) shows the estimated effect on birth cohort size. The positive interaction coefficient indicates that counties with greater rainfall deviation saw a larger number of survivors in the famine cohorts. Specifically, a one-unit increase in rainfall deviation was associated with an increase in famine cohort size by about 5%. These results support the hypothesis that weather shocks moderated famine severity.

Robustness. I conduct several further estimations, detailed in Supporting Information S1: A7, to test the robustness of the estimated effect of rainfall deviation on famine mortality. The estimate is robust to a variety of adjustments, including clustering the standard error at the prefecture level; removing observations after 1961; removing counties in Sichuan, the province that experienced the most severe famine; weighting the regression using the inverse of terrain ruggedness, as the accuracy of interpolated precipitation is lower in more rugged areas (Dell et al. 2014); using rainfall deviation observed one year earlier ($t-1$); calculating rainfall deviation using the long-term median or the historical average as the benchmark.[9]

Heterogeneities. To understand the impacts of different types of weather shocks, specifically floods and drought, I estimate the effect of rainfall deviation while considering its directions (i.e., positive and negative deviations). I categorize the observations

TABLE 3 | Weather shocks and famine mortality.

	Log death rate (1)	Log cohort size (2)
Rainfall deviation	0.009*	−0.008**
	(0.005)	(0.003)
Rainfall deviation × GLF period	−0.063***	0.048***
	(0.015)	(0.011)
Observations	20,335	21,361
Adjusted R^2	0.404	0.919
County FEs	Yes	Yes
Year FEs	Yes	Yes

Note: Dependent variables: *Death rate* measures the number of deaths per thousand population in each county-year. *Cohort size* measures the number of people born in a county-year based on sample census data. Standard errors clustered by county in parentheses.
***$p < 0.01$.
**$p < 0.05$.
*$p < 0.1$.

TABLE 4 | Weather shocks and grain retention.

	Grain retention per capita	
	Quantity (log) (1)	Below 165 kg (2)
Rainfall deviation	−0.069***	0.044***
	(0.007)	(0.007)
Rainfall deviation × GLF period	0.081***	−0.072***
	(0.015)	(0.019)
Observations	9392	9392
Adjusted R^2	0.746	0.439
County FEs	Yes	Yes
Year FEs	Yes	Yes

Note: Standard errors clustered by county in parentheses.
***$p < 0.01$.
**$p < 0.05$.
*$p < 0.1$.

into groups of equal size based on the value of rainfall deviation with specific directions. Then I estimate the effect of each rainfall deviation category on famine mortality by comparing them with the median category, which reflected the condition with minimal rainfall deviation.[10] The estimation suggests that both excessive rainfall (floods) and deficient rainfall (drought) were associated with a significant decrease in famine mortality (see Supporting Information S1: A8).[11]

I further examine the effect of rainfall deviation across counties with different staple crops. In counties where the staple crop's output was more sensitive to precipitation fluctuations, higher-level officials could have faced greater uncertainties when estimating grain output under abnormal precipitation. This may have resulted in more latitude for county officials to adjust procurement quotas. Conversely, for staple crops with more predictable yields under various weather conditions, the discretion for county officials to adjust procurement quotas may have been constrained. As such, we might observe heterogeneous effects of rainfall deviation associated with a county's crop specialization.

To measure a county's crop specialization, I use the percentage of total grain output contributed by wheat or rice in 1957. Wheat output is recognized to be more sensitive to precipitation fluctuations than rice. In counties where wheat was the staple crop, officials could have had more latitude to adjust procurement quotas in response to precipitation fluctuations, potentially resulting in greater decrease in famine mortality. This interpretation is supported by empirical results. Regression estimates show a larger effect of rainfall deviation on famine mortality in counties where wheat occupied a larger share of grain output. In contrast, the effect of rainfall deviation decreased as the share of rice in total grain output increases (see Supporting Information S1: A9).

The impact of weather shocks on famine mortality also varied with a county's geographic proximity to provincial capitals. For counties located closer to the capital, weather shocks had a relatively larger negative effect on famine mortality. This effect diminished with increasing distance to the provincial capital (see Supporting Information S1: A10). One possible explanation is that in the geographically remote counties, where monitoring capacity was already limited, weather shocks did not significantly alter bureaucratic discretion in grain extraction. Alternatively, counties near the provincial capital may have had greater political and economic leverage, allowing the county leaders to negotiate extraction quotas more effectively in response to adverse weather conditions.

Mechanism. The estimated effect of rainfall deviation on famine mortality could be explained by county officials' implementation of grain extraction orders. The information gap associated with weather shocks provided an opportunity for county officials to moderate state targets for grain extraction with lower risk of penalty. This moderation of state targets could benefit the local population by increasing grain retention in rural communities, which created a vital grain reserve for survival during the famine. I illustrate this mechanism by analyzing the impact of rainfall deviation on grain retention during the famine, following the model specification in Equation (1) with grain retention as outcome variables.

Table 4 presents the estimated effects of rainfall deviation on grain retention.[12] The estimates indicate a significant temporal change in the relationship between rainfall deviation and grain retention during the famine. While rainfall deviation led to a decline in grain retention in ordinary years, this negative impact reversed during the famine. Despite production declines, counties with greater rainfall deviation managed slightly higher retention levels in the famine years, compared with counties with lower rainfall deviation (Column 1). This pattern was present not only in the change in grain retention levels, but also in the probability of excessive grain extraction–annual retention per capita fell below 165 kg (Column 2). During the famine, rainfall deviation was associated with a lower likelihood of grain retention below this subsistence level. This change suggests that county officials exercised their discretion during the famine to secure food supply for local farmers, whose survival hung precariously on the level of grain retention. Such adjustments in grain procurement quotas might explain, as least partially, the lower famine mortality observed in counties with greater rainfall deviation.

I further test whether the change in grain retention due to weather shocks led to lower famine mortality using a two-stage estimation. I first regress grain retention on rainfall deviation and save the predicted outcome. Next, I estimate the effect of the predicted grain retention on famine mortality. The estimation shows a significant impact of predicted grain retention on famine mortality. As the grain retention level increases, famine mortality rates decreased significantly (see Supporting Information S1: A11). This result supports the assumption that grain retention operated as a key mechanism that drove the relationship between rainfall deviation and famine mortality.

Apart from state procurement, the CCP's grain extraction policy included another component that could have affected food availability in rural communities. Under certain circumstances, local officials were authorized to provide state-owned grain to farmers to address societal needs. I conduct another test examining whether rainfall deviation explained the variations in state provision of grain to farmers (*fanxiaoliang*), also known as "return sale." I collect data on the return sale of grain for counties in four provinces, using the quantity of state-provided grain divided by county population to measure the degree of return sale. I find that return sale per capita was positively correlated with rainfall deviation, but this correlation showed no significant change during the famine (see Supporting Information S1: A12). This result suggests that return sale, compared to grain procurement, might be a secondary mechanism through which the impact of rainfall deviation on famine mortality operated.

5.3 | Weather Shocks, Dialect Distance, and Famine Mortality

While weather shocks created an opportunity for county officials to exercise greater discretion in grain procurement, the actual adjustments to procurement quotas can vary depending on social proximity between county officials and the population they governed. Greater social proximity may have led county officials to prioritize citizens' well-being over state objectives when they conflicted with each other. In contrast, with low social proximity, county officials were more likely to adhere to state targets at the expense of social welfare. As such, when potential adjustments to the grain procurement quotas were made possible by weather shocks, the reduction in procurement is expected to be more pronounced when social proximity between county officials and the county population was greater. I test this by estimating the heterogeneous effects of rainfall deviation, examining how they varied depending on dialect distance between county party secretaries and the county population.

Figure 4 illustrates how dialect distance moderated the effects of rainfall deviation. The coefficients, estimated with a triple interaction, reveal how the effect of rainfall deviation varied with dialect distance during the famine. When dialect distance was low, rainfall deviation significantly reduced the probability of excessive grain extraction and mortality rates during the famine. These effects diminished and became statistically insignificant as dialect distance increases. These results suggest that when there was a lower linguistic and cultural divide between a county party secretary and the county population, the county party secretary was more likely to make discretionary adjustments to grain procurement quotas in response to societal needs during the famine. Such responsiveness, while only present among counties with socially embedded officials, was consequential for mitigating famine mortality.

I validate the result above using the conventional proxy for social proximity–the dummy variable for native county party secretaries. Consistent with the result shown in Figure 4, there was a significant negative effect of rainfall deviation on famine mortality when a county party secretary was native to the province they took office. For counties governed by non-native

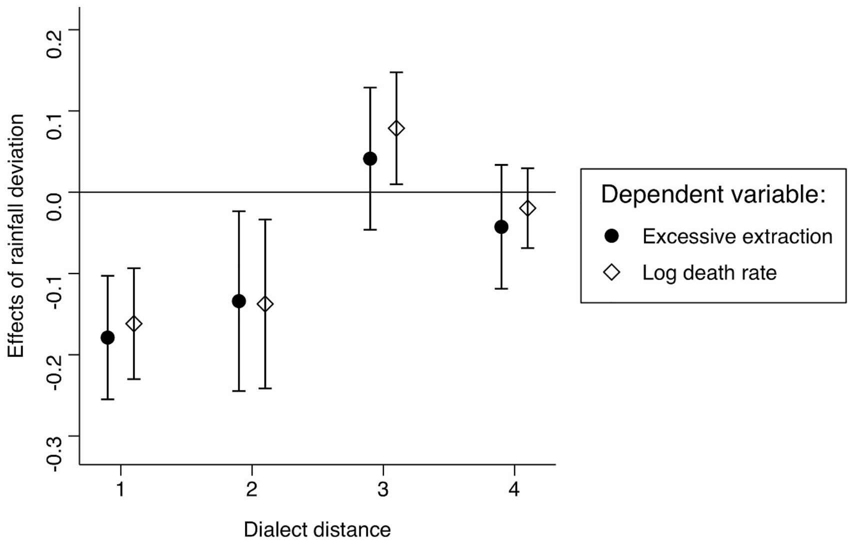

FIGURE 4 | Social proximity and heterogeneous effects of rainfall deviation. A coefficient represents an estimated effect of rainfall deviation on famine-period grain extraction or famine mortality, conditional on a certain level of dialect distance. 95% confidence intervals based on standard errors clustered by county.

party secretaries during the famine, I find no significant effect of rainfall deviation on famine mortality (see Supporting Information S1: A13).

6 | Conclusion

This study illustrated the social consequences of bureaucratic oversight, drawing on evidence from China under Mao. It shows that weather shocks, which widened central-local information gaps regarding local grain production, created opportunities for local officials to lessen the execution of excessive grain extraction orders during the GLF, thereby mitigating the social costs of the state's grain procurement policy. This finding contributes to our understanding of the role of bureaucrats in shaping government behavior during times of social crisis. It complements classical accounts of famine that emphasize the role of political accountability in preventing social disasters (Sen 1981, 1999). Deficiencies in government responsiveness do not necessarily stem from weak state control over bureaucrats. In contrast, they could be a result of strong political control over the bureaucracy under weak political accountability.

This study also contributes to our understanding of social embeddedness of bureaucrats. While socially embedded bureaucrats are often observed to be more responsive to local citizens, this article shows that the impact of social embeddedness depends on the degree of vertical control in the bureaucracy. This finding has implications for the debate over decentralization in developing countries. In contexts where policies are misaligned with the preferences of citizens, fostering connections between bureaucrats and citizens may protect citizens, as least partially, from the negative impact of perverse state policies.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] The optimal level refers to the elite's expected extraction when they have accurate information about local production levels.

[2] Alternative datasets, such as Matsuura and Willmott's weather data, are based on interpolation from a small number of data points–only 20 cross-sectional observations for the entire country (Kung 2022). This can result in limited variation and low accuracy for the interpolated data. The interpolated data used in this article, in contrast, are based on over 450 cross-sectional observations and exhibit significant variation.

[3] I use total county population as a proxy for rural population, as rural population data are unavailable for a significant portion of counties.

[4] The central government stipulated that for general urban adult residents (those with minimal labor activities), the grain ration should not exceed 27.5 jin per month, which was equivalent to an annual ration of 165 kg.

[5] Also see Supporting Information S1: A3 for an illustration of the variable construction.

[6] For computational efficiency, I use prefecture fixed effects instead of county fixed effects, which leads to faster convergence of the maximum likelihood estimation.

[7] Because output P is greater than county-wide retention R, it is straightforward that $0.05P - 0.05R = 0.05(P - R) > 0$. For a procurement rate at 50% (i.e., $(P - R)/P = 0.5$), the estimates suggest that the quantity of production decline would be twice as much as the quantity of retention decline.

[8] This is obtained from $(1 - e^{-0.063}) \times 100\% \approx 6\%$.

[9] The historical average precipitation for calendar year t ($1954 \leq t \leq 1965$) is calculated by the average precipitation between 1951 and year t.

[10] The median group corresponds to an interval of rainfall deviation centering around the long-term average and falling into the range $(-0.13, 0.1)$.

[11] In the case of extremely negative rainfall deviation, however, we observe a null effect on famine mortality. It was possible that those counties experienced production collapse so severe that the bureaucratic discretion could not meaningfully mitigate mortality.

[12] Because data on grain extraction are available for only a subset of counties, the effective sample size for the estimations is lower the optimal sample size.

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Supporting Information

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